



WeCloudData

Prompt Engineering

Introduction





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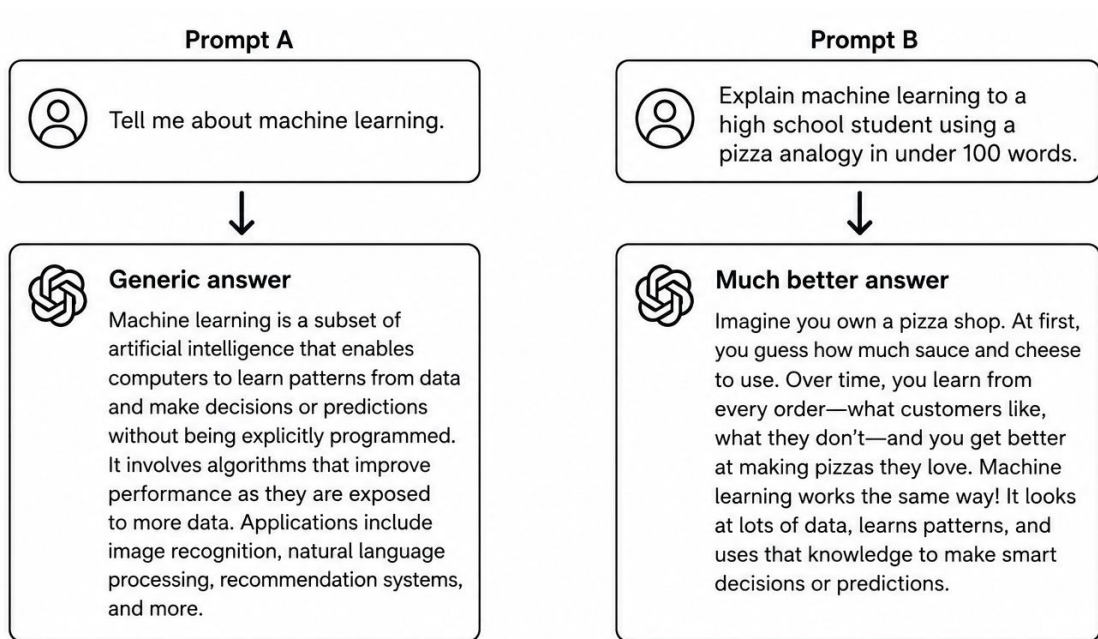
Introduction Prompt Engineering



Why Do We Need Prompt Engineering?

Introduction

Imagine asking GPT:

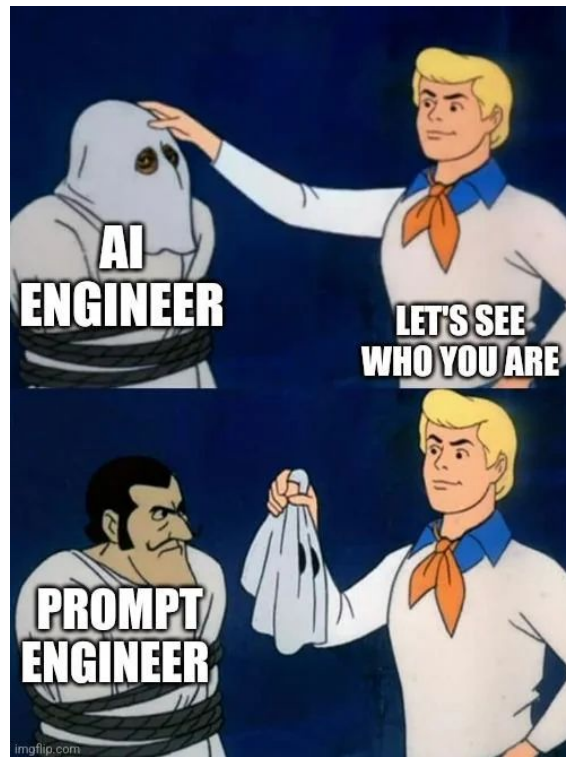




What is Prompt Engineering?

Introduction

Prompt Engineering is the practice of writing clear and effective instructions that help an AI model generate useful responses.





Why and When Use Prompt Engineering?

Introduction

Why?

- Improves response quality
- Reduces ambiguity
- Produces consistent outputs
- Makes AI easier to control

When?

Use Prompt Engineering whenever you interact with LLMs for

- Chatbots
- Coding
- Summarization
- Translation
- Content generation



How LLM Interpret Prompts:

Introduction

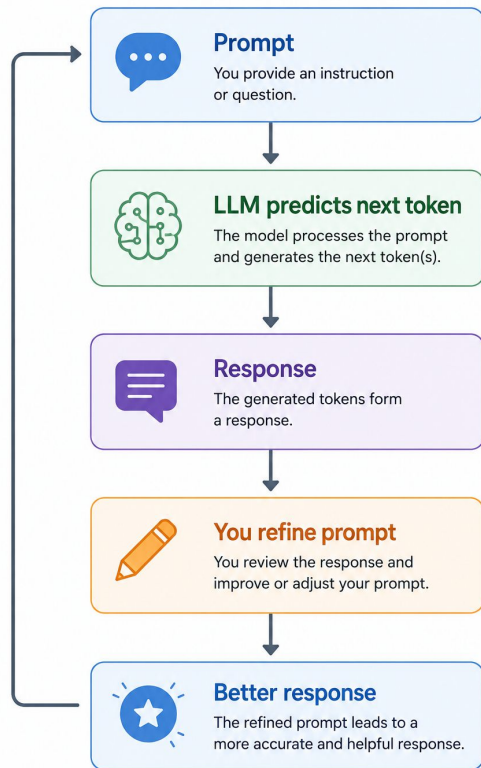
Instead of "thinking,"

LLMs predict the next most likely token.

Your prompt guides that prediction.

If prompts matter, what actually makes a prompt good?

Prompt Refinement Loop





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Anatomy of a good prompt

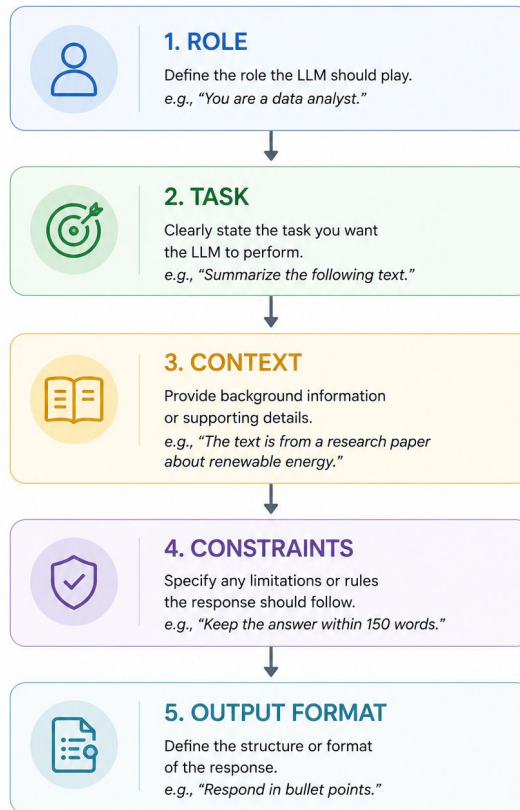
Prompt Engineering



What Makes a Good Prompt?

Anatomy of Good Prompt

Every strong prompt has five ingredients.





Example

Anatomy of a good prompt

Weak Prompt

Tell me how to improve my resume I'm applying for a data analyst job and I don't know what to write for skills also should I include my internship from 2 years ago and what if I don't have experience with SQL.

- The task, context, and questions are mixed together.
- The expected response structure is unclear.



Example

Anatomy of a good prompt

Labels

Task: Improve my resume

Context:

I am applying for a data analyst role.

Concerns:

- Skills section
- Internship from two years ago
- No SQL experience

Delimiters

```
<resume>
  <role>Data Analyst</role>
  <concerns>
    <skills>Not sure what to include</skills>
    <internship>Completed 2 years ago –
include?</internship>
    <sql>No experience – how to address?</sql>
  </concerns>
</resume>
```



Example

Anatomy of a good prompt

Lists /

I'm applying for a data analyst job. Please help me improve my resume by answering the following:

1. What technical and soft skills should I include?
2. Should I list an internship from 2 years ago?
3. I don't have SQL experience—how should I handle that?

Q&A

I'm applying for a DA role. Please answer the following questions to help me improve my resume:

- Q1:** What are the most important technical and soft skills I should include for a data analyst position?
- Q2:** I completed an internship two years ago. Should I still include it on my resume?
- Q3:** I don't have experience with SQL. How can I address this gap effectively?

Input/Output

Input:

I'm applying for a data analyst position. I need help improving my resume.

Output:

- Recommended skills to include
- Whether or not to list the internship
- How to address lack of SQL experience



Weak Vs Good Prompt

Anatomy of a good prompt

Weak Prompt



Example Prompt

Tell me about climate change.



Example Response

Climate change is a long-term change in Earth's temperatures and weather patterns. It is mainly caused by human activities like burning fossil fuels, which release greenhouse gases. It leads to melting ice caps, rising sea levels, and more extreme weather. People and governments are working to reduce emissions and transition to clean energy.

Good Prompt



Example Prompt

You are a tutor explaining climate change to high school students. Write a short summary (5 bullet points) covering: causes, key effects, and solutions. Use simple language and avoid technical jargon.



Example Response

- **Causes:** Mainly caused by human activities like burning fossil fuels, which release greenhouse gases.
- **Effects:** Melting ice caps, rising sea levels, extreme weather, and harm to plants and animals.
- **Impact on People:** Affects health, food supply, and communities, especially in vulnerable areas.
- **Solutions:** Use clean energy, save energy, plant trees, and reduce waste and emissions.
- **What You Can Do:** Save energy, use public transport, recycle, and spread awareness.



Weak Vs Good Prompt

Anatomy of a good prompt

A good prompt has:

Clear Task

Clearly state what you want the model to do

Separate Concerns

Keep different tasks separate in the prompt. One request at a time works best.

Relevant Context

Provide necessary background and details so the model understands the context

Defined output structure

Specify the format or structure of the output eg. json, bullet points etc.



Anatomy of a Good Prompt

Anatomy of a good prompt

Once we know how to write prompts, we can use different prompting strategies depending on the task.



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Prompting Techniques Prompt Engineering



Basic Prompting Techniques

Prompting Techniques

Technique	When to use?
Zero-shot	The task is simple and clear
One-shot	You need to show one expected example
Few-shot	You need to demonstrate a consistent pattern
Role-based	You want a specific tone or perspective
Chain-of-Thought	The task requires multiple steps



Zero-shot Prompting

Prompting Techniques

What?

Zero-shot prompting asks the model to perform a task **without providing any examples**.

The model relies only on its pre-trained knowledge and the instructions in the prompt.

When?

Use Zero-shot Prompting when:

- The task is simple and well-defined.
- The model is likely to already understand the task.
- You don't need a specific output style or format.



Zero-shot Prompting Example

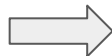
Prompting Techniques

Example: Sentiment Classification

Prompt

Classify the following review as positive, negative or neutral.

Review: The product works well but the delivery was very slow



Possible Output

Neutral





One-shot Prompting

Prompting Techniques

What?

One-shot prompting provides **one example** before asking the model to perform a similar task.

The example helps the model understand the expected output format or style.

When?

Use One-shot Prompting when:

- You want to demonstrate the expected format.
- The task is slightly more complex.
- A single example is enough to guide the model.



One-shot Prompting Example

Prompting Techniques

Example: Sentiment Classification

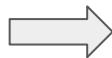
Prompt

Review: The setup was easy and the app works perfectly.

Label: Positive

Now classify:

Review: The product works well, but delivery was very slow.



Possible Output

Neutral





Few-shot Prompting

Prompting Techniques

What?

Few-shot prompting provides **multiple examples** before asking the model to solve a new task.

The examples help the model recognize patterns and improve consistency.

When?

Use Few-shot Prompting when:

- The task is more complex.
- You need a consistent output style.
- One example isn't enough to explain the pattern.



Few-shot Prompting Example

Prompting Techniques

Example: Sentiment Classification

Prompt

Review: Excellent quality and fast delivery.

Label: Positive

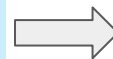
Review: It stopped working after one day.

Label: Negative

Review: The quality is good, but the packaging was damaged.

Label: Neutral

Now classify: **Review:** The product works well, but delivery was very slow.



Possible Output

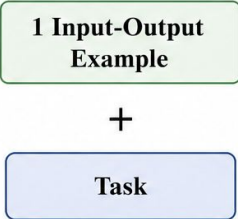
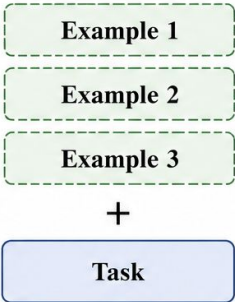
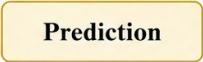
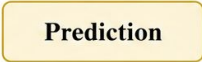
Neutral





Comparison

Prompting Techniques

	Zero-shot	One-shot	Few-shot
Setup (Input to Model)			
Flow			
Description	The model predicts the output using only the task instruction.	The model sees one example (input-output pair) plus the task, then predicts the output.	The model sees multiple examples (input-output pairs) plus the task, then predicts the output.



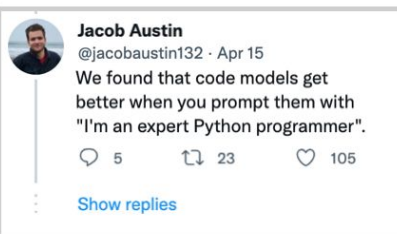
Comparison

Prompting Techniques

Few-shot learning

**Few-shot learning
+ instructions**

**Few-shot learning
+ instructions
+ intermediate steps**



https://x.com/_jasonwei/status/1516844920367054848





Role Based Prompting

Prompting Techniques

What?

You assign the model a role or persona to guide how it responds.

Effect:

Guides tone, perspective, vocabulary, and level of detail.

Connection:

A role is one optional component of a prompt.

Role-Based Prompting

```
You are a grizzled pirate captain tellin' yer crew how to find  
the hidden treasure on Skull Island. Speak in full pirate lingo,  
matey!
```



Role-based Prompting Example

Prompting Techniques

ROLE ✨	WHAT YOU GET ✨
 TEACHER Explains things step-by-step.	 Clear, simple, <u>easy to learn</u> 
 LEGAL ADVISOR Gives safe, formal, risk-aware advice.	 Structured, professional, <u>with disclaimers</u> 
 ACADEMIC RESEARCHER Provides deep analysis with evidence & citations.	 Rigorous, objective, <u>well-researched</u> 
 COPYWRITER Writes words that sell & convince.	 Persuasive, on-brand, <u>high-conversion</u> 



Chain of Thought Prompting

Prompting Techniques

What?

A prompting approach that asks the model to work through a multi-step problem carefully.

Chain-of-Thought (CoT)

You have a square pyramid with a base side length of 12 units and a height of 9 units. Find the total surface area of the pyramid, including the base. *Let's think step by step.*

CoT Model Response

Total Surface Area of Square Pyramid

- Base: $12^2 = 144$
- Slant height: $l = \sqrt{9^2 + 6^2} = \sqrt{117} \approx 10.82$
- Triangle area: $\frac{1}{2} \times 12 \times 10.82 = 64.92$
- Lateral area: $4 \times 64.92 = 259.68$
- Total area: $144 + 259.68 = 403.68$ units²



Choosing the right Technique

Prompting Techniques

Task	Best Technique
Simple Q&A	Zero-shot
Follow Format	One-shot
Learn Pattern	Few-shot
Expert Advice	Role based
Multi-step Reasoning	Chain of Thought



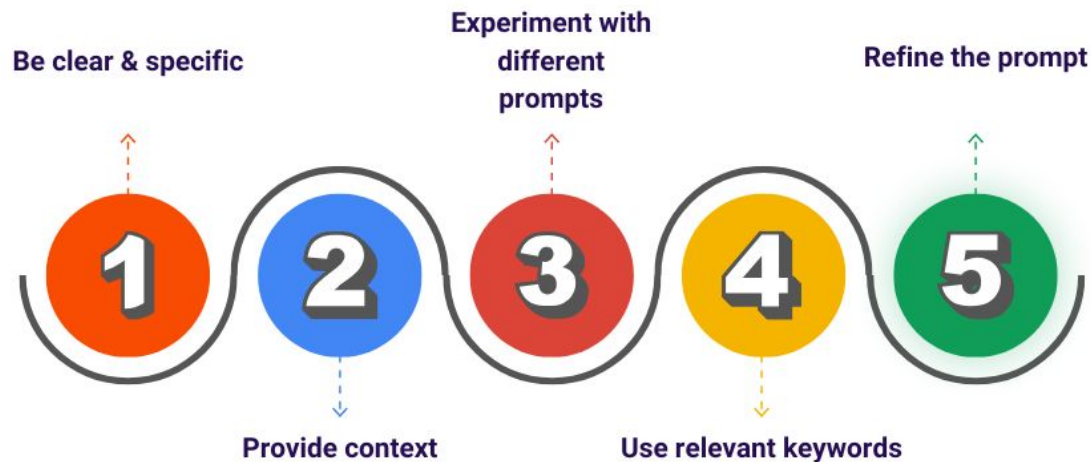
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Best Practices **Prompt Engineering**



Writing Better Prompts

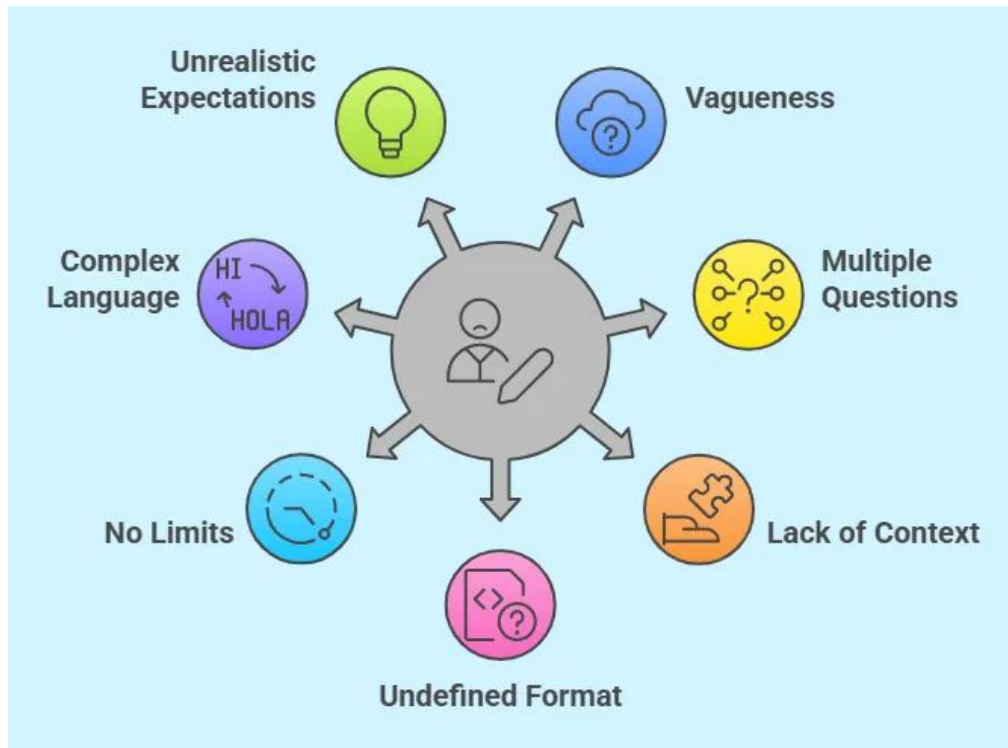
Best Practices





Common Mistakes

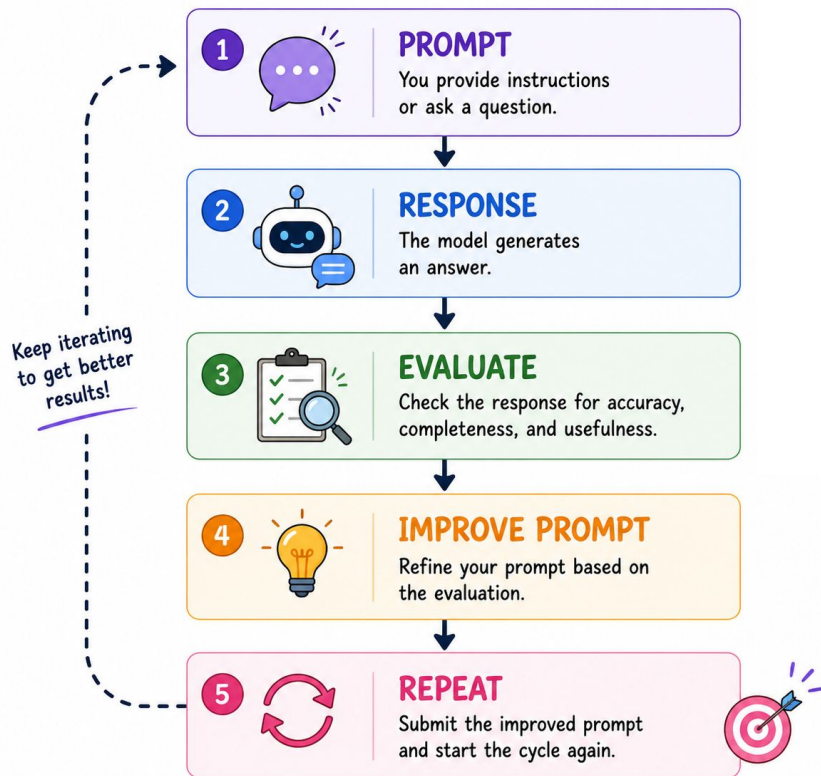
Best Practices





Prompt Refinement

Best Practices





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Prompt Engineering in Practice

Prompt Engineering



Common Prompting Tasks

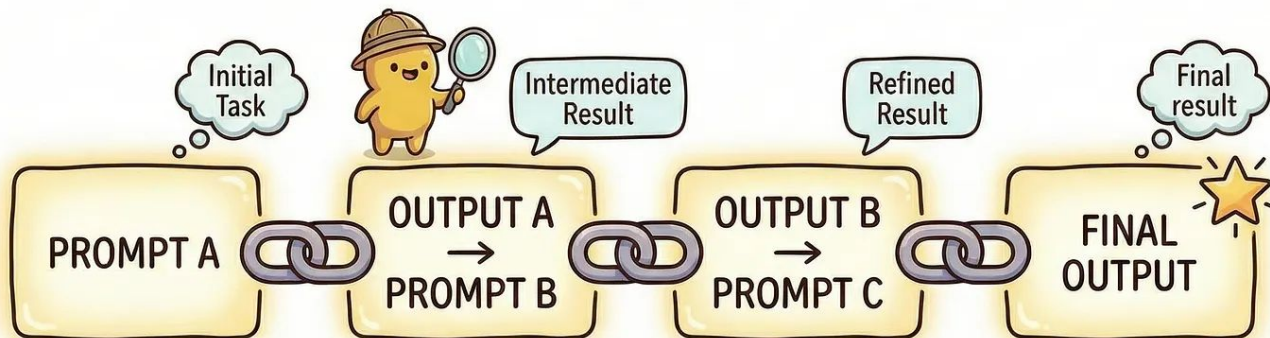
Prompt Engineering in Practice

Task	Prompt
Summarization	Summarize the following paragraph in 3-4 sentences capturing the key points
Translation	Translate the following text from English to Arabic
Classification	Classify the following email with “Spam” or “Not Spam”. Respond with one word only.
Coding	Write a python function to check if a number is prime or not prime.



Prompt Chaining

Prompt Engineering in Practice



Linear sequence

Step by step

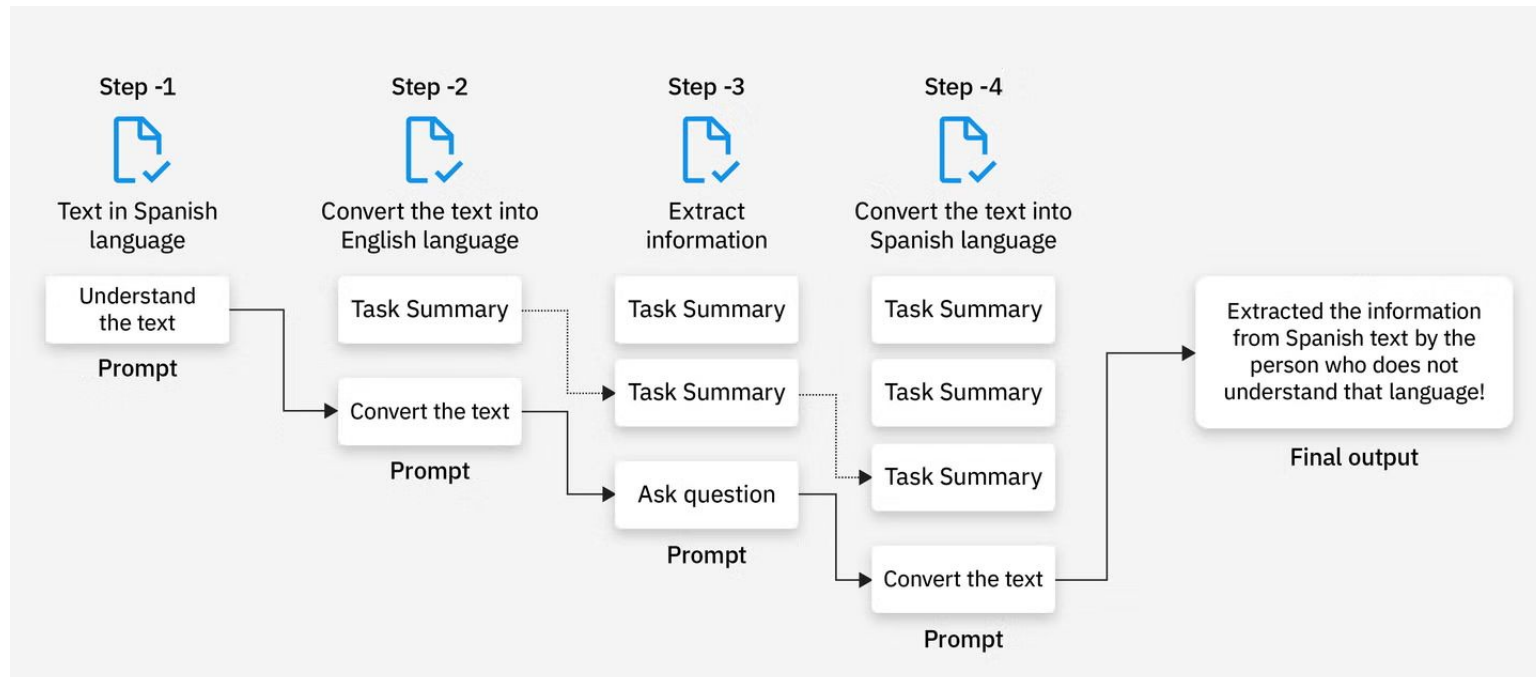
Predictable flow

Iterative refinement through connected prompts.



Prompt Chaining Example

Prompt Engineering in Practice





Summary

- What Prompt Engineering is
- Prompt anatomy
- Prompting techniques
- Best practices
- Real-world usage



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Thank You

We'd love to hear from you. If you have any questions,
feel free to reach out to our team anytime!